Q6. Gray Code → Binary

a) 01101 (gray)

0 1 1 0 1

0 1 0 0 1

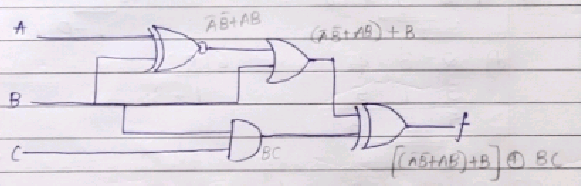
b) 1 0 1 0 1

1 1 0 0 1

Assignment - 2 - Boolean Algebra

1. Derive the output expression 'f' for the below circuits and draw the simplified circuits.

a)



$= (\overline{A}\overline{B}+AB)+B$

$\overline{A}\overline{B}+B(A+1)$

$\overline{A}\overline{B}+B$

$\underline{\overline{A}+B}$

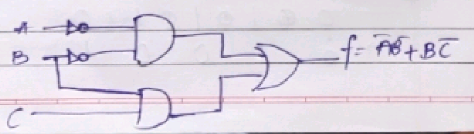
$f = (\overline{A}+B) \oplus BC$

$= (\overline{\overline{A}+B})BC + (\overline{A}+B)\overline{BC}$

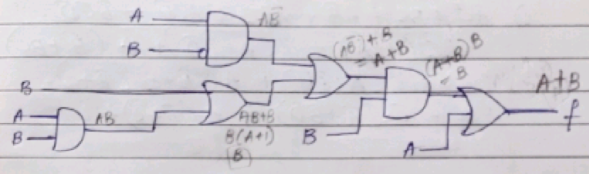
$= (A\overline{B})BC + (\overline{A}+B)\overline{BC}$

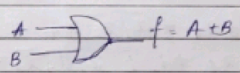
$=$

$f = \overline{A}\overline{B} + B\overline{C}$

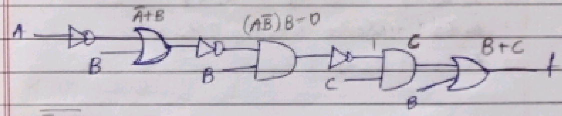


b)



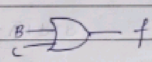


c)



$\overline{\overline{A}+B} = A\overline{B}$

$f = B + C$



2. Minimize the Boolean expressions

a) $\overline{A}C\overline{D}E + \overline{A}\overline{B}\overline{D} + ABCE + ABD$

$\overline{A}\overline{D}(CE+\overline{B}) + AB(CE+D)$

$\overline{A}\overline{D}((\overline{B}+C)(\overline{B}+E)) + AB(D+CE)$

$\underline{\overline{A}\overline{D}(CE+\overline{B})+AB(D+CE)}$

b) $\overline{A}\overline{B}\overline{C} + ABD + \overline{A}C + \overline{A}C\overline{D} + A\overline{C}D + A\overline{B}\overline{C}$

$= \overline{B}\overline{C}(\overline{A}+A) + \overline{A}(C+C\overline{D}) + AD(B+\overline{C})$

$= \overline{B}\overline{C} + \overline{A}(C(1+\overline{D})) + AD(B+\overline{C})$

$= \overline{B}\overline{C} + \overline{A}C + AD(B+\overline{C})$

c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + AC\bar{D} + \bar{B}\bar{C}\bar{D}$

$BC(1+\bar{A}) + \bar{C}(\bar{A}+\bar{B}\bar{D}) + A(\bar{B}+BC)$

$= BC + \bar{C}(\bar{A}+\bar{B}\bar{D}) + A(\bar{B}+B)(\bar{B}+C)$

$= BC + \bar{C}(\bar{A}+\bar{B}\bar{D}) + A(\bar{B}+C)$

d) $\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_1x_2\bar{x}_3x_4 + x_2$

$= x_2(x_1\bar{x}_3x_4 + 1) + \bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4$

$= \bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1x_4 + x_2$

$= \bar{x}_1\bar{x}_3 + x_3x_4 + \bar{x}_1x_4 + x_2$

$= \bar{x}_1\bar{x}_3 + x_3x_4 + x_2$

3. $Z = \bar{A}\bar{B}\bar{C} + AB\bar{C} + A\bar{B}C + \bar{A}BC$

$Z = \overline{A \oplus B \oplus C}$

the entire circuit is equivalent to a 3-input XNOR gate

4. Minimize the Boolean expressions using K-map.

a) $f(a,b,c,d) = \sum_m(4,6,8,10,11,12,15) + \sum_d(3,5,7,9)$

ab \ cd	00	01	11	10
00	0	1	X 3	2
01	1 4	X 5	X 7	1 6
11	1 12	13	1 15	14
10	1 8	X 9	1 11	1 10

$\Rightarrow \bar{a}b + a\bar{b} + a\bar{c}\bar{d} + \cancel{b\bar{c}d}\ cd$

b) $f(a,b,c,d) = \pi_M(0,2,4,6,7,9) \cdot \pi_d(10,11)$

ab \ cd	00	01	11	10
00	0 0	1	3	0 2
01	0 4	5	0 7	0 6
11	12	13	15	14
10	8	0 9	X 11	X 10

$\Rightarrow \cancel{\bar{a}\bar{d} + \bar{a}bc + a\bar{b}d}$

$(a+d)(a+\bar{b}+\bar{c})(\bar{a}+b+\bar{d})$